

TrES-3b

R-2587

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_101029 · created 2026-08-27T22:54:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455498.62926 +/- 0.00089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.029
Transit depth (Rp/Rs)^2	1.6 +/- 0.74 [%]
Orbital Inclination (inc)	82.81 +/- 1.44
Airmass coefficient 1 (a1)	1.283 +/- 0.017
Airmass coefficient 2 (a2)	-0.1185 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	None
Optimal Aperture	3.16
Optimal Annulus	8.61
Transit Duration (day)	0.057 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.127 ± 0.029 vs 0.16309 (deviation 1.21σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.02σ)
Mid-time O–C	0.6 min
Depth SNR	4.2
Residual scatter	1.33 %

Light curve

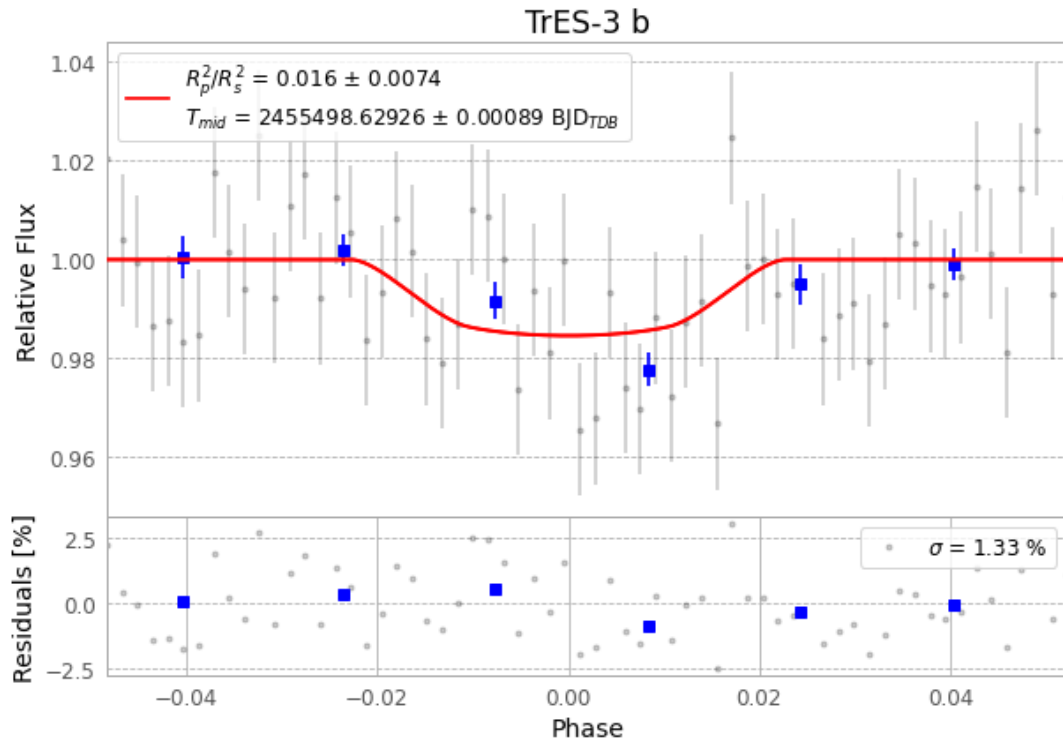


Figure 1 — FinalLightCurve_TrES-3 b_2010-10-28.png (SHA-256 dc63bfb9d38234be...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 190], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b1228e282e258be3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), TRES-3101029013614.FITS → TRES-3101029044514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-101029.log (8123d7f91d16...), FinalLightCurve_TrES-3 b_2010-10-28.png (dc63bfb9d382...), FinalLightCurve_TrES-3 b_2010-10-28.csv (ae5e11c643ad...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 22:54Z.